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Psychological research frequently is concerned with latent constructs that cannot be observed directly, such as intelligence or personality. Because these constructs are not directly measurable, they are inferred from manifest indicators, such as item responses or behavioral measures, using measurement models. Consequently, the validity of substantive conclusions depends fundamentally on the quality of the measurement process. If a measure does not adequately represent the construct of interest, even highest statistical rigor cannot prevent misleading conclusions (Flake and Fried 2020; Flake et al. 2017). However, recent work still has emphasized the need for stronger statistical methods (Flake et al. 2017; Haig 2020; Altoè et al. 2020) as well as more explicit theoretical foundations in psychology (Eronen and Bringmann 2021). Although these discussions often focus on issues of statistical inference or theory construction, both ultimately depend on the quality of measurement. Regardless of the sophistication of a statistical model or the strength of a theoretical framework, conclusions remain limited if the underlying measurement instrument does not function as intended.

A central aspect of measurement quality is measurement invariance (MI), which refers to the assumption that a measurement model operates equivalently across individuals, groups, contexts, or measurement occasions. Under MI, the relationship between a latent construct and its observed indicators remains stable across the conditions being compared (Meredith 1993). This assumption is essential whenever researchers seek to compare latent means, variances, covariances, or structural relations, making MI a fundamental prerequisite for many correlational, experimental, and longitudinal research designs (Putnick and Bornstein 2016).

Violations of MI imply that the relationship between the latent construct and its observed indicators differs across groups or conditions. In such cases, differences in observed responses may not solely reflect differences in the underlying construct but may also arise from systematic differences in item functioning or measurement precision. Consequently, observed group differences become ambiguous, as they may represent a combination of substantive construct differences and measurement-related distortions (Cheung and Rensvold 2002; Meredith 1993). Without establishing MI, it is therefore impossible to determine whether apparent differences reflect true variation in the latent construct or artifacts of the measurement process. Consequently, MI is not only a methodological requirement but also a prerequisite for valid theoretical interpretation.

Classical Measurement Invariance Testing

Structural equation models (SEMs) are widely used to define latent constructs and their relations to observed indicators. Within SEMs, measurement models specify the relationships between latent factors and their manifest indicators. By separating construct-related variance from measurement error, latent variable models provide a principled basis for evaluating measurement quality (Bollen 1989).

Classical approaches to measurement invariance testing are fundamentally confirmatory in nature. Researchers begin by specifying a measurement model, typically in the form of a confirmatory factor analysis (CFA), that reflects substantive hypotheses about the construct under investigation. A categorical grouping variable is then used to partition the sample into two or more groups, across which measurement parameters are constrained and compared. This most widely used approach for evaluating measurement invariance is usually referred to as multiple-group confirmatory factor analysis (MGCFA) (e.g. Cheung and Rensvold 2002; Putnick and Bornstein 2016).

In MGCFA, measurement invariance is typically assessed through a sequence of nested model comparisons in which equality constraints are imposed on increasingly restrictive sets of parameters. Equality of the models structure in both groups is called configural invariance, and can be understood as both groups share the same factor structure (i.e. number of latent factors and items). Equality of factor loadings former is referred to as metric invariance, if metric invariance is present it indicates that all factor loading parameters are equal across groups. Specifically, this suggests that the items of a given scale and their underlying constructs relate to each other with the same strengths in both groups. Scalar invariance further constrains item intercepts to equality across groups. Establishing scalar invariance is generally required for meaningful comparisons of latent means because it indicates that the measurement scale has the same unit and origin across groups. Residual or strict invariance assumes, additionally to the constraints above, that variables residuals are equal across the groups. If the residual variance holds, the items of the scale are measuring the latent constructs with the same measurement error. This implies that it can indeed be known whether differences measured by the items of the tested scale between the groups are due to group differences in the proposed factors or due to measurement errors (Cheung and Rensvold 2002; Widaman and Reise 1997). The level of invariance required depends on the substantive inference of interest. For example, scalar invariance is generally required when comparing latent means across groups, whereas less restrictive forms of invariance may be sufficient for other research questions (Putnick and Bornstein 2016).

For MGCFA testing, the assessment of measurement invariance depends on two central specifications: the measurement model itself and the grouping variable used to represent potential sources of heterogeneity. As a result, measurement noninvariance (MNI) can only be detected with respect to the grouping variables included in the analysis. Consequently, this framework is highly effective when substantive theory naturally implies discrete group distinctions, such as treatment conditions, demographic categories, or measurement occasions. However, many characteristics that may influence measurement functioning vary continuously across individuals. In such situations, representing heterogeneity through predefined groups may require arbitrary categorization and can obscure important variation in measurement parameters.

Moderated Nonlinear Factor Analysis

These considerations have motivated the development of approaches that conceptualize MNI as a continuous process operating at the individual level. One prominent example is the Moderated nonlinear factor analysis (MNLFA). MNLFA conceptualizes heterogeneity in data as functions of observed moderator variables. Within MNLFA, measurement parameters are allowed to vary systematically as functions of one or more moderator variables. Consequently, MNI is conceptualized as moderation of the measurement model itself. If measurement parameters differ as a function of the moderator, the measurement model no longer operates equivalently across individuals and therefore violates measurement invariance.

Whether MI is present depends on which parameters are moderated by the exogenous variable. The exogenous variable may influence any subset of the parameters in the measurement model. Specifically, the means, variances, and covariance of latent factors, as well as the item intercepts, factor loadings, and residual variances of any manifest indicator, may vary systematically across individuals as deterministic functions of the exogenous variable. If moderation is limited to the latent-factor parameters, that is, the means, variances, and covariance, the model remains consistent with MI. In contrast, moderation of item-level parameters, including intercepts, loadings, or residual variances, constitutes MNI. Accordingly, a moderation function must be specified for each parameter that is allowed to vary as a function of the exogenous variable. Because of this conceptualization,this method allows researchers to model gradual, nonlinear, and individual-specific forms of MNI that are difficult to represent within traditional multiple-group approaches.

However, this increased flexibility comes at the cost of greater specification requirements. Like MGCFA, moderated nonlinear factor analysis (MNLFA) constitutes a fundamentally confirmatory approach to the assessment of MI. In practice, at least three sets of decisions are required. First, the measurement model must be specified, including the latent structure and the relationships between latent and observed variables. Second, one or more moderator variables must be selected to represent potential sources of heterogeneity in the measurement process. Third, moderation functions must be specified for all parameters that are allowed to vary as a function of the moderator. This includes determining which parameters are moderated as well as specifying the functional form of each moderation effect (e.g., linear, quadratic, or another parametric form). Consequently, the detection of measurement non-invariance depends not only on the choice of moderator variables but also on the assumptions made about how these moderators influence the model parameters.

Structural Equation Model Trees

Structural Equation Model trees (SEM trees) were originally introduced by Brandmaier et al. (2013) as a method for identifying important covariates and their interactions within SEMs. Subsequent developments further refined the methodology and expanded its applicability (Arnold et al., 2021; Brandmaier et al., 2016). Notably, although measurement invariance was used as an illustrative application by Arnold et al. (2021), SEM Trees were not originally developed as a dedicated method for testing measurement invariance. Because violations of MI manifest as systematic heterogeneity in measurement parameters, SEM trees appear conceptually well suited for detecting such effects.

SEM trees build on two paradigms: SEMs and on decision trees. Decision trees describe recursive partitions of the covariate space, whereby the sample is repeatedly split according to covariate values into progressively more homogeneous subsets, in order to maximize differences in a given outcome variable. SEM Trees combine the hierarchical structure of the decision trees with SEMs, so that they describe heterogeneity in data with respect to a specified SEM as targeted outcome structure. Unlike many other tree-based methods, SEM Trees can be applied to models that include latent variables. Despite their conceptual suitability for this purpose, SEM trees have received relatively little attention as tools for detecting MNI.

In contrast to MGCFA and MNLFA, SEM trees represent a more exploratory approach to the assessment of MI. Similar to these methods, SEM trees require the researcher to specify a measurement model a priori. Beyond this, however, the researcher only needs to provide a set of candidate partitioning variables that may account for heterogeneity in the data. These variables may be numerous and can be either categorical or continuous. In this respect, the researcher decisions required by SEM trees closely resemble those of the traditional MGCFA framework.

Similar to MGCFA, researchers specify a measurement model and identify variables that may account for heterogeneity. The crucial difference is that MGCFA requires the researcher to select a specific grouping variable and define its manifestation in advance, whereas SEM trees can evaluate a large set of candidate variables simultaneously and determine which, if any, are associated with violations of MI. Moreover, SEM trees are able to identify nonlinear forms of moderation and data-driven partition points for continuous variables without requiring these relationships to be specified a priori.

Consequently, SEM trees retain the confirmatory component associated with the measurement model while adopting an exploratory approach to the identification of heterogeneity. This combination makes them particularly attractive for researchers who are confident in their measurement model but have limited theoretical guidance regarding the sources or structure of MNI. Because the required researcher decisions largely mirror those of the familiar MGCFA framework, the transition from traditional MI testing to SEM trees should involve a comparatively low methodological threshold. At the same time, SEM trees offer substantially greater flexibility by accommodating multiple candidate moderators, continuous and categorical partitioning variables, and potentially nonlinear forms of heterogeneity. Although explicit specification of moderators and moderation functions remains advantageous when strong theoretical expectations are available, SEM trees provide a complementary strategy for situations in which the sources or functional forms of heterogeneity are uncertain.

Present Study

More generally, MGCFA, MNLFA, and SEM trees can be understood as reflecting different assumptions regarding prior knowledge about measurement heterogeneity. MGCFA assumes that relevant sources of heterogeneity can be represented through predefined groups. MNLFA further assumes that researchers can identify relevant moderators and specify how these moderators influence model parameters. SEM trees, in contrast, place fewer assumptions on the structure of heterogeneity and instead emphasize data-driven identification of parameter differences. Consequently, the relative performance of these approaches is likely to depend on the extent to which researchers possess accurate prior knowledge about the sources and functional forms of MNI.

The present simulation study compares the performance of MNLFA and SEM trees in detecting MNI across a range of conditions that vary in sample size, reliability, effect magnitude, and the form of moderation. To reflect both the methodological capabilities of the investigated approaches and plausible empirical scenarios, the simulation incorporates different functional forms of MNI, including linear, quadratic, and sigmoid moderation processes. We expect SEM trees to perform particularly well in conditions characterized by nonlinear moderation, especially sigmoid relationships, whereas their relative advantage should diminish in purely linear settings. In contrast, MNLFA is expected to perform best when the functional form of moderation is correctly specified, particularly in linear conditions, but to be more sensitive to misspecification of the underlying moderation function. For both methods, stronger moderation effects are expected to increase the detectability of measurement non-invariance.

To enhance the study's relevance for empirical practice, the simulation is grounded in a realistic applied-research perspective. Specifically, we consider researchers whose experience with measurement invariance testing is rooted in the traditional MGCFA framework and who seek to adopt more flexible approaches for detecting heterogeneity. Accordingly, MNLFA and

SEM trees are evaluated not only as statistical procedures but also as methodological alternatives for researchers moving beyond conventional MGCFA-based invariance testing.

Method

The simulation study was conducted in two complementary studies that differ in the complexity of the underlying latent-variable model. “Study 1” focuses on MNI within a single-factor framework and investigates how the performance of MNLFA and SEM trees is affected by characteristics of the moderation process, including the number of moderated parameters, the strength and functional form of moderation, sample size, and indicator reliability. Particular emphasis is placed on conditions in which the analytical model may incorrectly represent the true moderation function. “Study 2” extends this framework to a two-factor model. In addition to measurement-level noninvariance, heterogeneity may also occur in relations among latent variables through moderation of the latent covariance. Furthermore, “Study 2” incorporates structural-model misspecifications, allowing the robustness of both methods to be evaluated under more complex latent-variable structures.

Data-Generation Mechanism

Across all simulation conditions, MNI was generated through moderation of model parameters by an observed continuous covariate. A raw moderator M was sampled from a uniform distribution on the interval $[-1, 1]$ and transformed into an effective moderator $Z = h(M)$. The transformation $h(\cdot)$ determined the functional form of heterogeneity and varied across simulation conditions. Importantly, nonlinearity was introduced exclusively through the transformation $Z = h(M)$, while moderation effects remained linear in the transformed moderator. Model parameters subject to moderation were specified as deterministic functions of Z, such that measurement properties changed systematically across individuals.

Functional Form of Moderation

The effective moderator $Z = h(M)$ was generated systematically using the following DGM functions:

1. Linear
2. Quadratic
3. Sigmoid
4. Noise/null moderation, $h(M) = 0$

These functions were selected to represent qualitatively different forms of heterogeneity. Linear moderation reflects the functional form most commonly assumed in modeling applications,implying a constant change in measurement parameters across the moderator range.

In contrast, quadratic moderation captures situations in which measurement differences are strongest at extreme values of a characteristic, such as when respondents at very low and very high levels of a trait interpret items differently than individuals near the center of the distribution. Sigmoid moderation reflects threshold-like processes in which measurement parameters change little across most of the moderator range but shift more rapidly once a critical region is reached, for example when item functioning remains stable until a particular level of symptom severity, cognitive ability, or developmental maturation is exceeded. The null condition ($h(M) = 0$) served as a measurement-invariant baseline and was used to evaluate Type I error rates.

Strength of Moderation

In each case where parameters are subjected to moderation, the strength of the moderation δ is varied:

δ_λ is evaluated at the following discrete levels:

$$\delta_\lambda \in \{-0.3,\ -0.2,\ 0.2,\ 0.3\}.$$

$$\delta_\nu \in \{-1.0,\ -0.5,\ 0.5,\ 1.0\} \quad \text{for } i \in \{1,2\}, \qquad \delta_\nu = 0 \quad \text{for } i \in \{3,4\},$$

Model Misspecification of Moderator Form

The DGM always defines the effective moderator through $Z = h(M)$. In MNLFA, the analytical model is fitted using prespecified parametric forms, namely linear and quadratic moderation. Consequently, MNLFA is correctly specified when the fitted analytical form corresponds to the DGM transformation and misspecified otherwise. SEM trees do not impose a parametric functional form, but they must identify the correct moderator variable(s) from the candidate set. The simulation assumes that the moderators are always perfectly measured. So this really reflects an ideal scenario and should be kept in mind, when interpreting the results. We expect that one should see a similar picture, but with less-than-perfect reliabilities and lower power in general, should one add variation to the reliability of the moderators.

Performance Criteria

The primary estimands in this simulation study pertain to the detection and accurate estimation of MNI due to moderation effects on measurement parameters. Specifically, we assess whether the estimation methods compared can correctly identify the presence or absence of moderation in factor loadings, intercepts, and residual variances, reflecting the true data-generating structure. Consequently, key simulation outcomes include Type I error rates and statistical power associated with detecting such moderation effects, as well as bias and variability in the recovered parameter estimates. Secondary estimands include model-level indicators of fit (e.g., Kullback–Leibler divergence, AIC), which serve as auxiliary diagnostics for methods that yield such metrics (i.e., MNLFA). These are not to the same extent applicable to tree-based approaches and are thus interpreted as method-specific targets rather than general evaluative criteria.

For the parametric MNLFA, we will extract estimated moderated parameters (e.g., factor loadings, intercepts), their associated standard errors, and two-sided p-values for testing the null hypothesis of no moderation effect. Additionally, model fit indices such as

Kullback–Leibler divergence, AIC and the LRT-statistics will be recorded. For the parameter estimates and fit indices, we will summarize their distributions across simulation replications by reporting key quantiles (2.5th, 50th, and 97.5th percentiles). The null hypothesis will be rejected if the associated p-value is less than the conventional significance level of 0.05. In contrast, non-parametric tree-based approaches do not yield direct parameter estimates with standard errors. We use score-based SEM-trees (Arnold et al. 2021). Their detection of MI/ non-invariance is operationalized via the presence of at least one split in the correct condition at the correct respective level of focus parameters. Those splits, introducing a subgroup, will be evaluated at the significance level: 0.05. Additionally, a Bonferroni correction to account for multiple comparisons at each node. The trees will be allowed to split on the focus parameters relevant for the invariance testing (the means of the intercepts, factor loadings and the latent covariance in study 2). Candidate subgroup structures are assessed by testing whether they produce statistically significant differences in the specified model parameters across groups. Differences in factor variances between subgroups are permitted in the model, but they are not used as criteria for selecting or evaluating covariates. Furthermore, pre-pruning will be applied, limiting the maximum depth of the tree to 3 and min.**N** is set to 50. Because the MNLFA also directly considers the form of the moderation, which the SEM trees do not, we further want to use SEM forests for generating partial dependence plots to get an approximation of the functional form of the moderating variables.

Replications

We employ a full grid search, in which parameter values are sampled from pre-specified discrete values. To ensure computational feasibility however, we followed an iterative approach whereby a initial pilot simulations $n = 100$ was conducted to amongst others, estimate the needed simulation time. These estimates then informed the total number of XX replications.

Software and Implementation

All simulations and analyses will be conducted in R (R Core Team 2025). The MNLFA-style models will be implemented directly in `OpenMx`(Boker et al. 2026), using raw-data maximum likelihood estimation and definition-variable-style moderation through OpenMx algebra objects. SEM tree analyses will be performed with the `semtree` (Brandmaier et al. 2026) package using OpenMx RAM models. Simulation design construction, data handling, aggregation, and visualization will rely on `dplyr`(Wickham et al. 2026), `tidyr` (Wickham et al. 2025), `tibble` (Müller and Wickham 2026), `purrr` (Wickham and Henry 2026), `furrr` (Vaughan and Dancho 2022), `future` (Bengtsson 2021), `parallel` (R Core Team 2025), and `ggplot2` (Wickham 2016). The simulation study will be conducted in R (R Core Team 2025).

General Analysis Procedures

Study 1

Population Models

Simulation Conditions

Focuses on functional form misspecification and data-related conditions:

- **Sample sizes:** $N \in \{300, 500, 700, 1,000\}$
- **Intercepts:** Intercept baselines are set to 1 across simulation conditions.
- **Indicator reliabilities:** *Low* (.6), *moderate* (.7), *moderate to high* (.8), or *high* (.95).

Results: Study 1

Study 2

Population Models

Simulation Conditions

Builds on Study 1 and introduces additional model misspecification in the factor structure:

- **Structural model misspecifications:** (between-subjects factor)
  - **1. No misspecification:** Factor structure matches the analysis model.
  - **2. Cross-loadings:** Items y_5, y_6 additionally load on a second latent factor (See Figure 3).
- **Latent Covariance:** The latent covariance is moderated by Z. The baseline variance is set to 1 and moderation strength is selected from:

$$\delta_{\sigma_\eta} \in \{-0.3, -0.2, 0.2, 0.3\}.$$

Results: Study 2

Discussion

Practical Recommendations

References

Altoè, Gianmarco, Giulia Bertoldo, Claudio Zandonella Callegher, et al. 2020. "Enhancing Statistical Inference in Psychological Research via Prospective and Retrospective Design Analysis." *Frontiers in Psychology*, advance online publication. https://doi.org/10.3389/fpsyg.2019.02893.

Arnold, M., M. C. Voelkle, and A. M. Brandmaier. 2021. "Score-Guided Structural Equation Model Trees." *Frontiers in Psychology* 11: 3913. https://doi.org/10.3389/fpsyg.2020.564403.

Bengtsson, Henrik. 2021. "A Unifying Framework for Parallel and Distributed Processing in r Using Futures." *The R Journal* 13 (2): 208–27. https://doi.org/10.32614/RJ-2021-048.

Boker, Steven M., Michael C. Neale, Hermine H. Maes, et al. 2026. *OpenMx: Extended Structural Equation Modelling*. https://doi.org/10.32614/CRAN.package.OpenMx.

Bollen, K. A. 1989. *Structural Equations with Latent Variables.* John Wiley & Sons. https://doi.org/10.1002/9781118619179.

Brandmaier, A. M., John J. Prindle, Manuel Arnold, and Caspar J. Van Lissa. 2026. *Semtree: Recursive Partitioning for Structural Equation Models.* https://github.com/brandmaier/semtree.

Cheung, Gordon W., and Roger B. Rensvold. 2002. "Evaluating Goodness-of-Fit Indexes for Testing Measurement Invariance." *Structural Equation Modeling: A Multidisciplinary Journal* 9 (2): 233–55. https://doi.org/10.1207/S15328007SEM0902/_5.

Eronen, Markus I., and Laura F. Bringmann. 2021. "The Theory Crisis in Psychology: How to Move Forward." *Perspectives on Psychological Science* 16 (4): 779–88. https://doi.org/10.1177/1745691620970586.

Flake, Jessica K., and Eiko I. Fried. 2020. "Measurement Schmeasurement: Questionable Measurement Practices and How to Avoid Them." *Advances in Methods and Practices in Psychological Science* 3 (4): 456–65. https://doi.org/10.1177/2515245920952393.

Flake, Jessica K., Jolynn Pek, and Eric Hehman. 2017. "Construct Validation in Social and Personality Research: Current Practice and Recommendations." *Social Psychological and Personality Science* 8 (4): 370–78. https://doi.org/10.1177/1948550617693063.

Haig, Brian D. 2020. "What Can Psychology's Statistics Reformers Learn from the Error-Statistical Perspective?" *Methods in Psychology* 2: 100020. https://doi.org/https://doi.org/10.1016/j.metip.2020.100020.

Meredith, William. 1993. "Measurement Invariance, Factor Analysis and Factorial Invariance." *Psychometrika* 58 (4): 525–43. https://doi.org/10.1007/BF02294825.

Müller, Kirill, and Hadley Wickham. 2026. *Tibble: Simple Data Frames.* https://doi.org/10.32614/CRAN.package.tibble.

Putnick, Diane L., and Marc H. Bornstein. 2016. "Measurement Invariance Conventions and Reporting: The State of the Art and Future Directions for Psychological Research." *Developmental Review* 41: 71–90. https://doi.org/https://doi.org/10.1016/j.dr.2016.06.004.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing.* R Foundation for Statistical Computing. https://www.R-project.org/.

Vaughan, Davis, and Matt Dancho. 2022. *Furrr: Apply Mapping Functions in Parallel Using Futures.* https://doi.org/10.32614/CRAN.package.furrr.

Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis.* Springer-Verlag New York. https://ggplot2.tidyverse.org.

Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2026. *Dplyr: A Grammar of Data Manipulation.* https://doi.org/10.32614/CRAN.package.dplyr.

Wickham, Hadley, and Lionel Henry. 2026. *Purrr: Functional Programming Tools.* https://doi.org/10.32614/CRAN.package.purrr.

Wickham, Hadley, Davis Vaughan, and Maximilian Girlich. 2025. *Tidyr: Tidy Messy Data*. https://doi.org/10.32614/CRAN.package.tidyr.

Widaman, Keith F., and Steven P. Reise. 1997. "Exploring the Measurement Invariance of Psychological Instruments: Applications in the Substance Use Domain." In *The Science of Prevention: Methodological Advances from Alcohol and Substance Abuse Research.* American Psychological Association. https://doi.org/10.1037/10222-009.